Research Methods Leading to a Perception of Knowledge Loss—One Century of Plant Use Documentation Among the Chácobo in Bolivia

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The loss of traditional knowledge, concomitant with changes in livelihoods, languages, and demographics of indigenous and local groups, is a global concern. However, documenting such loss poses serious methodological challenges. Comparing the results of contemporary studies with past research is often problematic due to methodological differences. Here, comparing studies that attempted to document the traditional ethnobotanical knowledge of the Chácobo of Bolivia, we tried to examine whether knowledge loss was really occurring across more than 100 years or was only researcher's perception. The Chácobo are a Panoan-speaking tribe of about 1000 members, first visited by researchers in 1911, and subsequently in the 1950s, 1960s, 1970s, 1980s, and 1990s. Each study had different foci, but all recorded ethnobotanical data. The first more detailed anthropological report exists from the late 1960s, and ecological-ethnobotanical studies were conducted in the 1980s and 1990s. Based on available literature, in particular the botanical studies of Boom (1987) and Bergeron (1998), it seemed that Chácobo plant use now centers on income generation. Both Boom (1987) and Bergeron (1998) perceived that traditional plant use related to household artifacts and medicine, as well as traditional crop varieties had almost disappeared. Here, we hypothesized that plant knowledge documented and the perception of so-called knowledge loss observed in these depended completely on the background of the interviewers and the methods employed, and that in a sufficiently comprehensive ethnobotanical study, we would be able to document all species and uses mentioned in previous studies. We tested this hypothesis by conducting a complete ethnobotanical

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inventory of almost the entire adult Chácobo population, with interviews and plant collection conducted directly by Chácobo counterparts. The results verify our initial hypothesis and showed that the loss of knowledge perceived in previous studies simply was an artifact of the research methods employed. Traditional crop varieties are still widely grown, most Chácobo know, and can name, traditional artifacts, and many still know the names and uses of medicinal species. However, some knowledge, including the manufacture of artifacts and proficient identification of many medicinal plants, is limited to the older generation.

La pérdida del conocimiento tradicional, los cambios en los medios de subsistencia, la pérdida de las lenguas locales, y la reducción demográfíca de los grupos indígenas y locales, es una preocupación mundial. Sin embargo, documentar dicha pérdida plantea serios desafíos metodológicos. Comparar los resultados de estudios recientes con investigaciones pasadas, no resulta fácil debido a las diferencias metodológicas. Aquí comparamos estudios que documentaron el conocimiento tradicional etnobotánico de los Chácobo en Bolivia, buscando examinar si la pérdida de conocimiento tradicional realmente ha estado sucediendo durante los últimos 100 años, o si solo era la percepción de los investigadores. Los Chácobo son una tribu Pano hablantes, actualmente conformada por aproximadamente 1000 miembros. Fueron visitados por primera vez por investigadores en 1911, y posteriormente en los años 1950, 1960, 1970, 1980 y 1990. Cada estudio tenía enfoques diferentes, pero todos registraron datos etnobotánicos. El primer informe antropológico más detallado existe desde fines de la década de 1960, y se llevaron a cabo estudios ecológico-etnobotánicos en los años ochenta y noventa. Basado en la literatura disponible, en particular los estudios botánicos de Boom (1987) y Bergeron (1998), la percepción fue que el uso de plantas por los Chácobo ahora se centraba en la generación de ingresos. Ambos autores percibieron que el uso de las plantas tradicionales relacionado con la fabricación de artefactos, la medicina tradicional, así como las variedades de cultivos tradicionales, casi habían desaparecido. Nosotros planteamos la hipótesis de que el conocimiento etnobotánico documentado y la percepción de pérdida de conocimiento observada en estudios previos, depende completamente de los antecedentes de los entrevistadores y los métodos empleados, sugiriendo que con un estudio etnobotánico suficientemente completo podríamos ser capaces de documentar todas las especies y usos mencionados en estudios previos. Probamos esta hipótesis realizando un inventario etnobotánico completo entrevistando casi la totalidad de la población adulta de los Chácobo y realizando la recolección de plantas, ambas actividades fueron desarrolladas directamente por contrapartes Chácobo. Los resultados verifican nuestra hipótesis inicial y muestran que la pérdida de conocimiento percibida en estudios previos simplemente fue un artefacto de los métodos de investigación empleados. Las variedades de cultivos tradicionales todavía se cultivan ampliamente; la mayoría de los Chácobo conocen y pueden nombrar los artefactos tradicionales, y la muchos aún conocen los nombres y los usos de las especies de plantas medicinales. Sin embargo, cierto tipo de conocimiento, incluida la fabricación de ciertos artefactos y la identificación de algunas plantas medicinales, se limitan a las generaciones más viejas.

Key Words: Knowledge loss, research methods, historic research, traditional knowledge.

Resumen: Pérdida de conocimiento, métodos de investigación, investigación histórica, conocimiento tradicional.

Introduction

Bolivia has a rich diversity of indigenous cultures, with at least 30 indigenous groups of 11 language groups. There are, however, few detailed ethnobotanical studies available. The Chácobo tribe in northeastern Bolivia is no exception. The first outsider to visit the Chácobo was the European anthropologist Erland Nordenskiöld in 1911 (Nordenskiöld 1922), followed by an anthropologist only in 1956, who published the last account of Chácobo life before the tribe came under the influence of American Evangelist missionaries (Haenke 1958). The Summer Institute of Linguistics (SIL) worked with Chácobo communities from 1953 to 1980 and produced the first account of Chácobo linguistics

(Prost 1960), and an unpublished work on Chácobo customs, with a strong focus on evangelist development (Prost 1970). This account is in interesting juxtaposition to the writings of German anthropologist Kelm (1972), who visited the Chácobo in 1970 and indicated that at this time the Chácobo were literally under SIL rule, with the missionaries trying to control all aspects of Chácobo life and trying to restrict any access to the tribe by outsiders. The SIL finally was replaced in 1980 by the Swiss Protestant mission. Missionary rule led to a profound change of lifestyle, and a permanent process of cultural change (Córdoba 2012). From 1983 to 84, Brian Boom (New York Botanical Garden) led the first ethnobotanical study of Chácobo, documenting their knowledge after almost 30 years of cultural change (Boom 1987). Boom did however base his work on the plants collected from a single 1-ha forest plot, with all collected species used as prompts for semi-structured interviews. In 1995, the Institut Francais d'Etudes Andines financed a re-survey of Boom's plot, employing the same methodology, but the results were never released to the public, and a planned publication (Bergeron 1998) existed in a single volume in the Institute's main library in Lima. Muñoz et al. (2000) published a study on anti-malarial plants used by the Chácobo, based on the results of Bergeron (1998). While the differences in methodologies do of course limit comparisons, the availability of previous studies on the Chácobo does still provide an interesting possibility to assess traditional knowledge (TK) over time.

TK has been recognized for its importance for the protection of ecosystem services and biodiversity (Huntington 2000; Shackeroff and Campbell 2007). However, researchers and policy-makers have equally expressed concern about its possible loss as societies modernize. A growing number of studies have reported changes and losses in TK (e.g., medical TK—Begossi et al. 2002; Case et al. 2005; Lozada et al. 2006; Monteiro et al. 2006, nutritional TK—Turner and Turner 2008, and agricultural TK—Benz et al. 2000; Stone 2007; Gómez-Baggethun et al. 2010). The hypothesis that TK systems are able to adapt to external changes and internal pressures has been discussed for some time (e.g., Berkes et al. 2000), leading to focus on the balance of conservative and generative processes within TK (McCarter and Gavin 2015; Quave and Pieroni 2015; Quave and Saitta 2016; Hart and Salick 2017). This dynamic nature of TK makes it an important part of the adaptive capacity of many rural and indigenous communities to balance biodiversity conservation and livelihood enhancement while adapting to disturbance and change (Colding et al. 2003; Berkes et al. 2000).

Few studies have however examined whether and how long-term loss or alteration of TK in fact occurs (Benz et al. 2000; Sternberg et al. 2001; Godoy et al. 2005; Reyes-García et al. 2005; McCarter and Gavin 2015; Quave and Saitta 2016). Consequently, our understanding of the resilience of TK systems and its ability to evolve and adapt is highly limited. The assumption of loss of TK, when younger people know less, is one of the common errors in ethnobotany (Hanazaki et al. 2013). This problem can be solved when analyzing the effects of age and age cohorts separately (Godoy et al. 2009; Reyes-García et al. 2013a). Not all TK domains might however be shared between generations, and as such, there can be domains that would be more vulnerable to TK loss and domains in which new knowledge is generated as an adaptation to environmental change (Reyes-García et al. 2013b; Paniagua-Zambrana et al. 2017).

Here, we hypothesized that plant knowledge documented and perceived knowledge loss observed in studies spanning 100 years covaried with the background of the interviewers and the methods employed, and that in a sufficiently comprehensive ethnobotanical study, we would see no, or very little, evidence of aggregate knowledge loss during this period of time.

Material and Methods

The Study Area—the Chácobo

The Chácobo belong to the Panoan linguistic group, which includes about 12 tribes (Chácobo, Pacahuara, Matis, Matses, Yaminahua, Ese Eja, and others). At the end of the 1890s, the Chácobo lived as semi-nomadic hunters and cassava and maize cultivators, probably in two groups, one with six families and one with four, in north Bolivia, between Lake Roguagnado and the river Mamore, south of their current territory (Boom 1987). During the rubber boom in the early 1900s, they were forced by more aggressive tribes to move north, where rubber tappers, who also brought disease and epidemics to the tribe, threatened them. While other tribes were enslaved to work in rubber stations, the Chácobo managed to avoid most of the outside influences. Their first permanent contact

with the outside world occurred only in 1953 with members of the New Tribes Mission, and in 1954, the Bolivian government established an agency about 15 km from the current location of Puerto Limones (Boom 1987). The missionary linguist Gilbert Prost arrived in 1955 under the auspices of the SIL. According to Prost (1970), there were four Chácobo groups living between the Benicito and Yata rivers at that time, numbering about 200 people (Boom 1987). Prost and his wife continued to live among the Chácobo until 1980. In addition to translating the New Testament into Chácobo, they made some observations on cultural and linguistic practices (Prost 1960, 1970). In 1964, Prost managed to buy a territory in the north of the Chácobo's ancestral lands, forming the community of Alto Ivón, and most of the remaining population moved there. In 1965, the Bolivian government finally assigned 43,000 ha of land to the Chácobo, although this area was less than 10% of their original territory. The influence of the SIL caused profound cultural change among the Chácobo, including the supposed abandonment of traditional costume and dance in 1969 (Prost 1970).

Currently, the population of the Chácobo community numbers about 1000 people (about 350 adults according to Central Indígena de la Región Amazónica de Bolivia—CIRABO pres. comm.), with Alto Ivón as the largest settlement. The current territory of the tribe encompasses 450,000 ha and is roughly equivalent to the original extent of the tribe's ancestral lands (Central Indígena de la Región Amazónica de Bolivia—CIRABO pres. comm.) (Fig. 1). The elevation of the territory is about 200 m, and much of the vegetation can be classified as humid tropical Amazon rainforest. However, the territory encompasses also large tracts of periodically inundated savannas, dominated by *Mauritiella armata* (Mart.) Burret, and large, drier, savanna areas with forest islands. The average annual temperature is 26.8 °C, with an average annual rainfall of 1560 mm. A distinct dry season lasts from June to November (Boom 1987). Today, the Chácobo are governed by two indigenous organizations: the Capitanía Mayor Chácobo, closely linked to the missionaries, and the Chácobo-Pacahuara Association, recognized by the Central Indígena de la Región Amazónica de Bolivia (CIRABO) and supported by the Central de Pueblos Indígenas del Beni (CPIB) and the Confederación de Pueblos Indígenas de Bolivia (CIDOB). Both organizations are however recognized by each other and by the Chácobo population.

Ethnobotanical and Botanical Collection

Our study explored the current TK of plant use of the Chácobo and Pacahuara in Beni, Bolivia and had three goals: (1) to discover and document current traditional plant knowledge through interviews and surveys, (2) to inventory the useful flora of the Chácobo, and (3) to repatriate the acquired knowledge as well as previous data to the community. Details on the study can be found in Paniagua-Zambrana et al. (2017).

After obtaining consent from CIRABO and before starting fieldwork, we conducted a community meeting in May 2013, involving representatives of all 27 villages in the Chácobo Territory, in order to obtain free prior informed consent from all communities. This session included the repatriation of the results of previous studies (Paniagua-Zambrana et al. 2011). In addition, during the project, all available material on Chácobo plant use was translated to Spanish and repatriated (Paniagua-Zambrana et al. 2014a). During this process, we extracted ethnobotanical information from previously published studies (Nordenskiöld 1922; Haenke 1958; Prost 1970; Kelm 1972; Boom 1987; Bergeron 1998). As prior studies often lacked demographic information, we could not consider age categories for this comparative analysis, but rather the aggregate set of knowledge presented in the individual studies.

For our comprehensive study, the assembly of the Chácobo community members itself choose ten local counterparts (two women, eight men, all of whom are listed as co-authors of this paper) to be trained as ethnobotanical interviewers and plant collectors. In September 2013, we conducted a 2-week workshop on ethnobiological methods and plant collection, training the selected counterparts. Training was conducted directly in the field in the central village of Alto Ivón and involved theoretical exercises (overview on methodology of interviews, collection, and herbarium techniques), as well as extensive practical exercises (structuring and testing of questionnaires, test interviews among the participants, field interviews with local community members, plant collection in the field, preparation of herbarium specimens, plant and artifact collection in the local community, data-basing, and initial data analysis).

From November 2013 to May 2015, Chácobo interviewers collected ethnobotanical information from 301 Chácobo participants (150 women, 151 men, representing almost the entire adult (> 18 years) Chácobo population), and over 1500 plant samples were collected. Prior to starting the

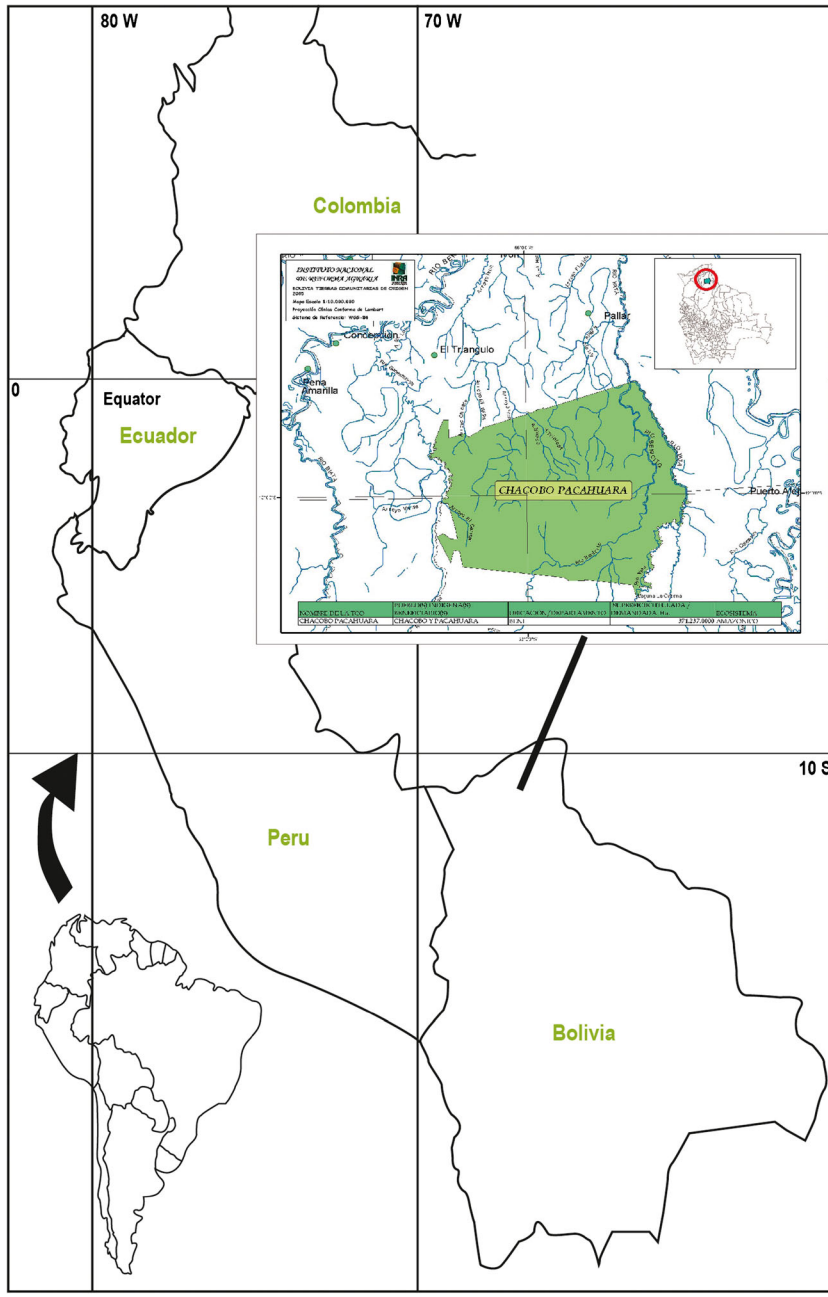

Fig. 1. Chácobo territory 2013.

interviews, every interviewer obtained free prior oral informed consent from each participant. Because the study attempted to interview the whole adult Chácobo population, there was originally no emphasis on achieving a balanced age or gender distribution. All interviews were conducted at the homes of the participants having the participants freelist their plant knowledge. All interviews were preferably conducted in Chácobo language by the trained Chácobo interviewers. In the few cases

where participants were not fully fluent in Chácobo, interviewers used Spanish as common language.

Plant material was identified and deposited at the National Herbarium of Bolivia (LPB) under the collection numbers of the Chácobo collectors. Nomenclature follows www.TROPICOS.org. For both prior and the current study, all uses and useful species reported in the interviews were classified in ten use categories following the Economic Botany Data Collection Standard (Cook 1995) with some modifications proposed by Macía et al. (2011).

A detailed description on the methodology is given in Paniagua-Zambrana et al. (2017).

All work was carried out following the International Society for Ethnobiology Code of Ethics (ISE 2006), and under the framework provided by of the "Nagoya Protocol on Access to Genetic Resources and Fair and equitable sharing of benefits arising from their use of the Convention on Biological Diversity," the Chácobo community retains the copyright of the TK of all informants. Any commercial use of any of the information requires prior consensus with informants and communities and an agreement on the distribution of benefits.

Data Analysis

Among studies, we compared plant species reported as a proportional Euler diagram, where circle areas and intersections approximate the counts and shared species for each area (Wilkinson 2011). In addition to these raw counts, we also used Euclidian distance as a metric of the difference among studies, comparing matrices in which studies were rows and plant genera (because a great number of plant species were unique to studies) were columns. We used the R package vegan (Oksanen et al. 2016) to construct ordinations based on these distances. We employed similar methods with the plant parts, Chácobo names, and use categories reported from each study. In the resulting ordinations, studies which report similar identities/abundances of items (genera, parts, uses, names) have a lower Euclidian distance and appear closer to one another and to the items that form the space in which they plot. Given the large number of unique species to each study, in order to highlight some species that might be especially prominent among these lists of unique species, we calculated indicator species (Dufrene and Legendre 1997), as implemented in the R package labdsv (Roberts 2016) for each study to combine occurrence frequency and mean abundance (number of pant part/use-category mentions).

Results

In the present study ("NATGEO"), we found overall 331 useful plant species, with 324 Chácobo names, in nine use categories with 45 subcategories, and detailed information on 15 plant parts used (Paniagua-Zambrana et al. 2017). Because our study involved a very large number of participants and focused on the whole flora, this surpassed the data found by all six previous studies (Nordenskiöld 1922; Haenke 1958; Prost 1970; Kelm 1972; Boom 1987; Bergeron 1998) on all counts (Table 1). NATGEO included almost all species mentioned as useful in previous studies (Fig. 2). While all species mentioned in the older studies were also found in our present research (see Paniagua-Zambrana et al. 2017 for a complete species list), the two large botanical studies (Boom 1987 and Bergeron 1998) did contain a variety of unique species, although only Bergeron (1998) did yield some unique indicator species, based on weighing species by numbers of mentions (Table 2). This can be explained by the fact that both botanical studies used the same 1-ha plot and employed a complete set of vouchers of this plot as prompts in semi-structured interviews, however based only on a

Table 1. Summary statistics across each of the eight studies.

Study	Species	Chácobo names	Use categories	Use subcategories	Parts used
Nordenskiöld 1922	20	2	6	11	6
Haenke 1958	9	1	3	5	4
Prost 1970	39	9	6	13	8
Kelm 1972	51	47	8	16	9
Boom 1987	399	317	8	37	11
Bergeron 1998	329	288	8	29	9
NATGEO	331	324	9	45	15

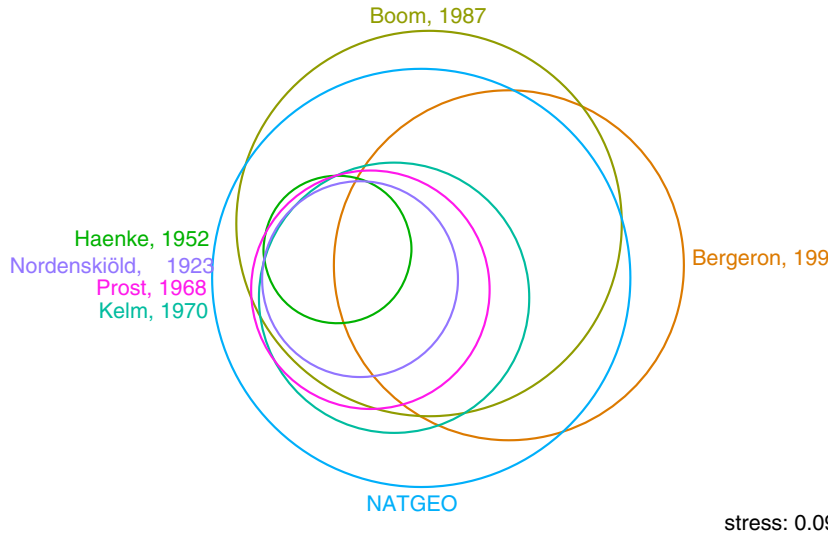

Fig. 2. Proportional Euler diagram, where circle areas and intersections approximate the counts and shared species for each area (Wilkinson 2011).

handful of participants in each study. Most genera (108) were however shared by the three big general studies (Boom 1987; Bergeron 1998; NATGEO). The two oldest studies (Nordenskiöld 1922; Haenke 1958) both focused their interviews especially on cultivated species (Fig. 3). Subsequently, most of the previous studies tended to report more fruit uses, while the present study elucidated on average more uses of plant trunks, especially for medicinal purposes (Fig. 4). Overall, every single use that had been mentioned in previous studies, including uses that had been regarded as discontinued by the botanical studies (Boom 1987; Bergeron 1998) (e.g., bark-cloth, a wide variety of hunting arrows, household utensils), was encountered in the present study, save one very specialized mention—a club, used for traditional dances and registered by Nordenskiöld (1922), did not appear in any of the later studies.

In compliance with the Nagoya Protocol, the original field notebooks, as well as the complete dataset, and a guide on useful plants of the Chácobo were repatriated to the Chácobo (Paniagua-Zambrana and Bussmann 2017). All members of the tribe have access to the compiled interview data for purposes of learning and education.

Discussion

The present study used the unique opportunity of the availability of multiple layers of previous field

TABLE 2. UNIQUE AND INDICATOR SPECIES FOR EACH STUDY.

Study	Unique species	Indicator species
Nordenskjoeld 1911	0	–
Haenke 1952	0	–
Prost 1968	0	–
Kelm 1970	0	–
Boom 1985	153 sp.	–
Bergeron 1995	131 sp.	*Roupala montana*, *Byrsonima chrysophylla*, *Miconia lourteigiana*, *Panopsis rubescens*
NATGEO	91 sp.	*Vismia macrophylla*, *Attalea phalerata*, *Apuleia leiocarpa*, *Cecropia ficifolia*, *Costus scaber*, *Oenocarpus bataua*, *Persea americana*, *Triplaris americana*

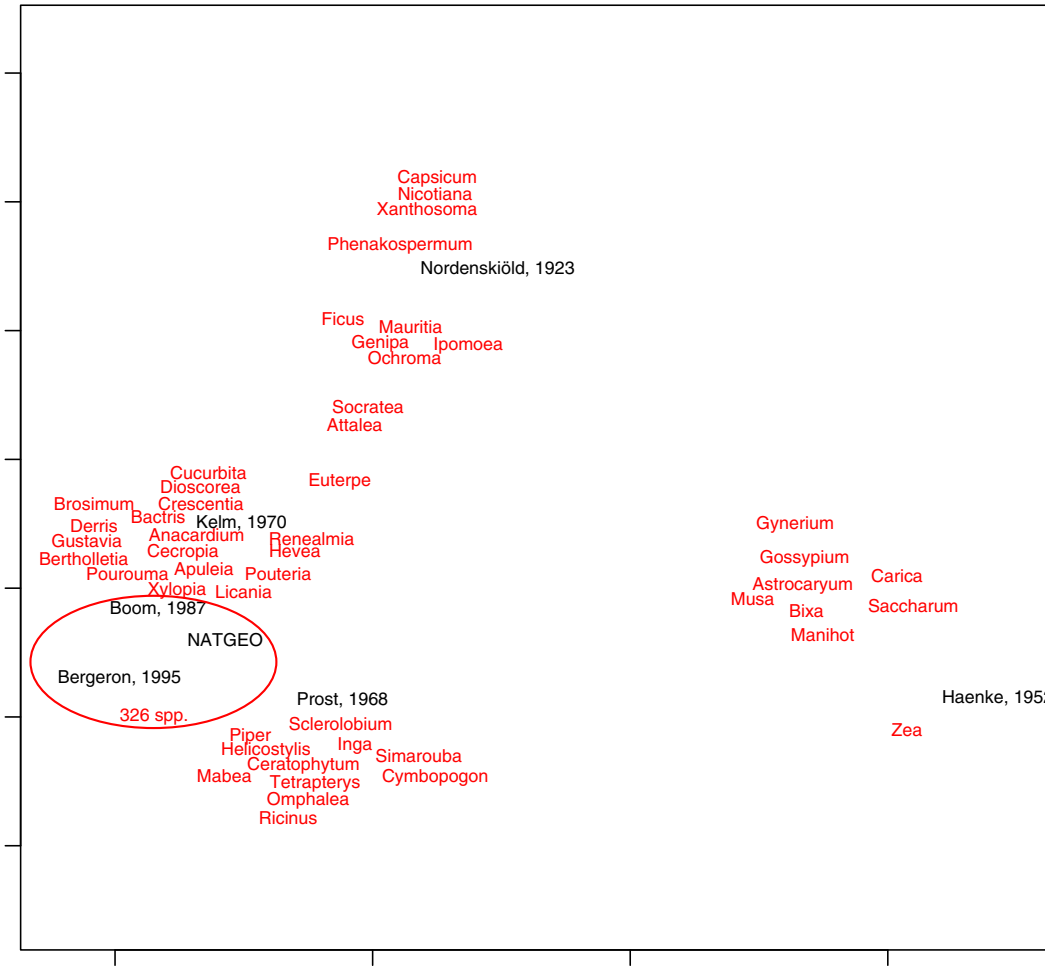

Fig. 3. Ordination with studies are plotted according to their similarities in plant genera reported.

data, although each was obtained using different technologies and focusing on different aspects of plant use, to shed light on whether the long-term loss or change in knowledge of an indigenous group over 100 years perceived in previous studies, was in fact occurring or was simply an effect of researcher's perceptions and methodology.

Many studies have however found patterns related to loss of knowledge attributed to different factors, such as decrease in plant diversity due to land use change (Shanley and Rosa 2004; Rocha and Cavalcante 2006), erosion of cultural practices and local languages (Benz et al. 2000), replacement of traditional education by formal schooling (Reyes-García et al. 2010), and economic factors that lead to migration, urbanization, new transportation routes, and integration into market economies (Godoy et al. 2005; Reyes-García et al. 2005). The observed changes in Chácobo life during the time of the analyzed studies (1911–2013) could clearly be related to the existence of some of these factors. The present freelisting study, to our knowledge the only available study involving almost the whole adult population of a tribe, revealed that indeed the perceived loss of knowledge among the Chácobo, as indicated particularly in previous botanical studies (Boom 1987; Bergeron 1998), was simply an artifact of research methodology.

Often, studies estimating cultural change have compared TK among different age cohorts or birth periods. The general trend found in relation to the difference in intergenerational knowledge, where

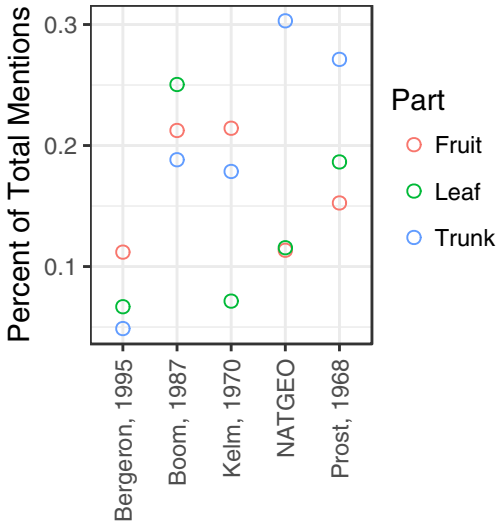

Fig. 4. Percent of total use mentions for each of the six largest studies that reported uses of specific plant parts.

older people know more than young people, suggests that this pattern generally could be a result of both knowledge transmission, as well as in situ learning, and be related to the time during which people acquire and use knowledge, with the older informants taking more responsibility in their households, who have a need to learn and apply their knowledge (Phillips and Gentry 1993; Godoy et al. 2009; Paniagua-Zambrana et al. 2014a; Zarger 2002). In addition, the trend of a gradual decrease in knowledge from the old to the very young has been related to increased exposure of the new generations to new technology and services (e.g., pharmacies, paramedics, clinics, and different types of building materials, tools, and alternative utensils available in commercial centers) (Byron 2003; Case et al. 2005; Godoy et al. 2009; Gómez-Baggethun and Reyes-García 2013; Paniagua-Zambrana et al. 2016; Reyes-García et al. 2013a, b; Vandebroek et al. 2004; Zarger 2002). Our recent study showed however that traditional plant knowledge among the Chácobo did not indicate any patterns related to age (Paniagua-Zambrana et al. 2017). Given that we interviewed almost the entire adult population (over 300 of about 350 adult Chácobo), we could prove that the knowledge was amply distributed among all generations (Paniagua-Zambrana et al. 2017). Some observed trends have also been explained by the adaptive nature of TK in response to livelihood changes (Ross 2002; Reyes-García et al. 2005, 2013a; Zent and Maffi 2010). The adaptive characteristic of cultural systems is based on the statement that beneficial environmental information passes from generation to generation, creating what is known as cumulative reservoirs of adaptive information, where human populations store the best efforts of the minds of previous generations (Boyd and Richerson 2006; Heinrich and McElrath 2003). An example of this TK property can be found in the difference between knowledge and actual use, as people do not necessarily use all resources stored in memory. According to this distinction, the group of species referred to as known, but not used by a community, is designated as "stock knowledge." The species that are used are recognized as "mass knowledge" (Albuquerque 2006). From an evolutionary viewpoint, the stock knowledge may be a strategy of local populations to ensure the existence of alternatives that can help them meet their needs if some resource is extinguished locally.

Orthodox thinking in anthropology and in economics predicts that local knowledge will vanish as economic development unfolds, but research suggests that this need not always be the case. Some researchers found that integration into the market through the sale of crops and wage labor is associated with less knowledge of wildlife, but integration into the market through the sale of forest goods is associated with more knowledge of wildlife (Godoy et al. 1998; Paniagua-Zambrana et al. 2014b). Others found no changes in the amount of local ecological knowledge a population had, over time, despite significant socioeconomic changes in the region (Zarger and Stepp 2004). However, in contrast, other studies confirmed the hypothesis that people who are relatively isolated from the market economy share more TK than people who live close to cities or larger towns (e.g., Reyes-García et al. 2005). Researchers have postulated that some resources may no longer be used due to industrialization or acquisition of technology services such as changes from wood to gas stoves and a replacement of medicinal plants by modern medicine (Baum and Myers 2004; Hanazaki et al. 2013). Although evident changes in Chácobo lifestyle occurred over the last century and the influence of social and economic change contributed to change, we found that the main part of the tribal knowledge documented in previous studies over the last 100 years is still present.

Our study has shown that in particular research focus, research methods, number of participants, times in the field, as well as the training and

ideological background of the interviewers greatly influenced the results obtained from the same tribal group over more than a century. The difference in methodology in particular could well be the main factor causing the perception of the knowledge loss perceived in previous studies, in particular Boom (1987) and Bergeron (1998).

Given the substantial knowledge able to be documented by training local interviewers and trying to interview almost the entire adult population in freelist interviews, we recommend that future studies should broadly train and involve local interviewers and a sufficiently large number of research participants, in order to fully document TK before speculating on knowledge distribution or knowledge loss.

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Author Contributions

NYPZ and RBU designed the study; NYPZ, RBU, ALHM, GOS, MOV, DOA, JSM, MSM, SC, BCM, GCM, and ES conducted the fieldwork; ALMH curated and identified the collections and entered the original data; NYPZ and RBU analyzed the data and NYPZ, RBU, and NPZ wrote the manuscript; REH conducted the statistical analysis; all authors read, corrected, and approved the manuscript.

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Compliance with Ethical Standards

Competing Interests. The authors declare that they have no competing interest.

Consent for Publication. This manuscript does not contain any individual person's data and further consent for publication is not required.

Ethics Approval and Consent to Participate. Before conducting interviews, both the permission of CIRABO and individual prior informed consent were obtained from all participants. No further ethics approval was required. All work conducted was carried out under the stipulations of the Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization to the Convention on Biological Diversity. The right to use and authorship of any traditional knowledge of all participants are maintained, and any use of this information, other than for scientific publication, does require additional prior consent of the traditional owners, as well as a consensus on access to benefits resulting from subsequent use.

Availability of Data and Materials. The raw data contain the names of all participants and cannot be shared publicly. Data without participant data can be obtained upon request after an access and benefit sharing agreement with CIRABO.

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